

Chapter 5

TOWARDS GREEN ENERGY



1. Remake the table taking into account relation between entries in three columns.

I	II	III
Coal	Potential energy	Wind electricity plant
Uranium	Kinetic Energy	Hydro electric plant
Water Reservoir	Nuclear Energy	Thermal plant
Wind	Thermal Energy	Nuclear power plant

I	II	III
Coal	Thermal Energy	Thermal plant
Uranium	Nuclear Energy	Nuclear power plant
Water Reservoir	Potential Energy	Hydro electric plant
Wind	Kinetic Energy	Wind electricity plant

2. Which fuel is used in thermal power plant? What are the problems associated with this type of power generation?

In the thermal power plant **coal** is used as a fuel.

Problems created by thermal power plant:

- i. Air pollution due to burning of coal: Burning of coal results in emission of gases like carbon dioxide, sulphur oxide and nitrogen oxide which are harmful to the health.
- ii. Along with the emission of gases due to burning of coal, soot particles are also released into the environment. This may cause serious health problems related to the respiratory system.

- iii. The reserves of fuel used in this method i.e., coal are limited. Therefore, in future, there will be limitations on the availability of the coal.

3. Other than thermal power plant, which power plants use thermal energy for power generation? In what different ways is the thermal energy obtained?

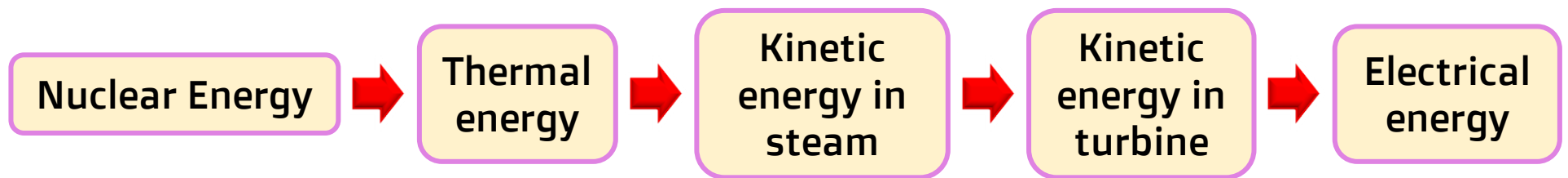
Other than thermal power plant, nuclear power plants, solar power plants use thermal energy for power generation.

Different ways of generating thermal energy are:

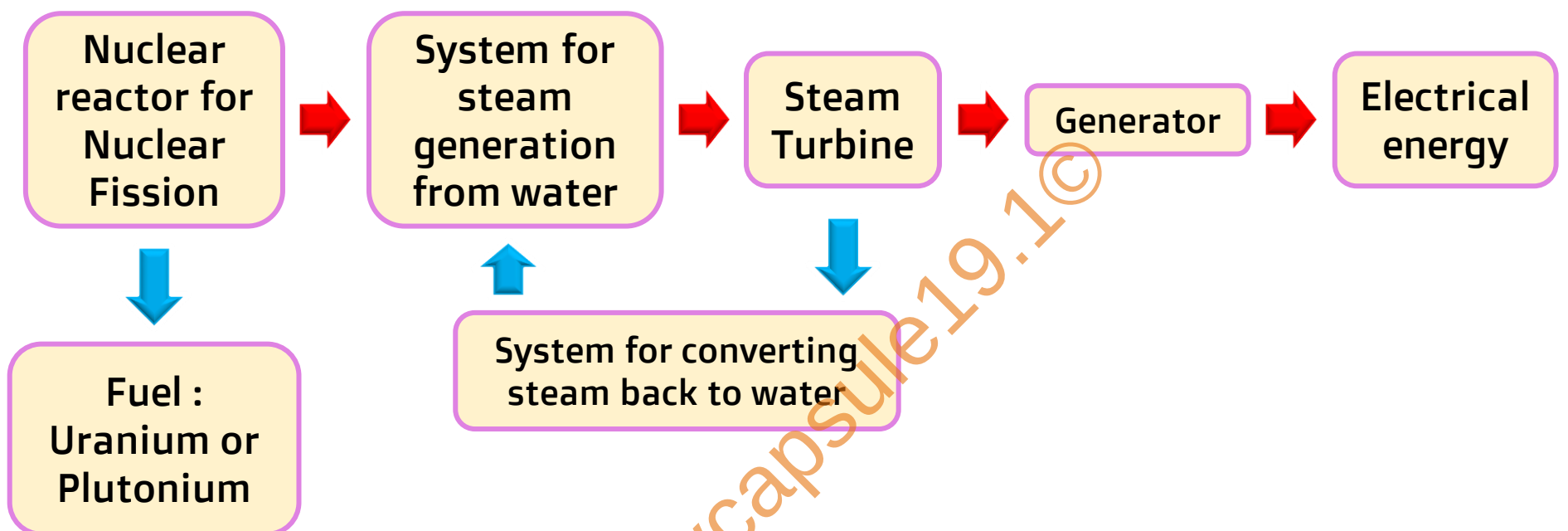
- i. Using fossil fuels: By burning fossil fuels such as coal, petroleum, the chemical energy stored in them is converted into thermal energy.
- ii. Using natural gas: Chemical energy in natural gas is converted into thermal energy.
- iii. Using nuclear fuels: In the nuclear power plant the nuclear fission of uranium results in the production of heat which is utilized to produce steam.
- iv. Using solar energy: In the solar thermal plant, the heat is obtained by concentrating solar radiations and then steam is produced.

4. Which type/types of power generation involve maximum number of steps of energy conversion? In which power generation is the number minimum?

- i. The nuclear power plant has maximum number of steps for the generation of electricity.



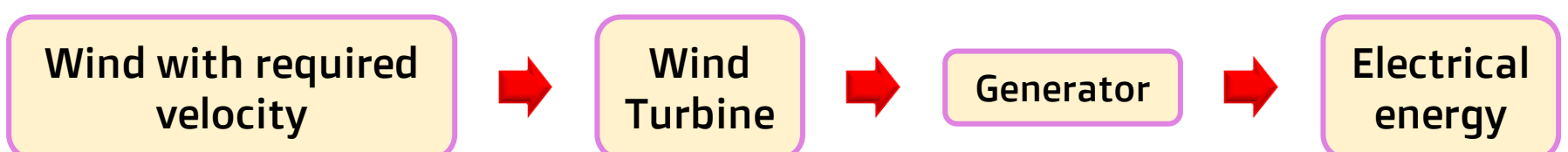
- ii. The flow chart of nuclear power plant:



- iii. The transformation of energy in electricity generation using wind energy has minimum number of steps.

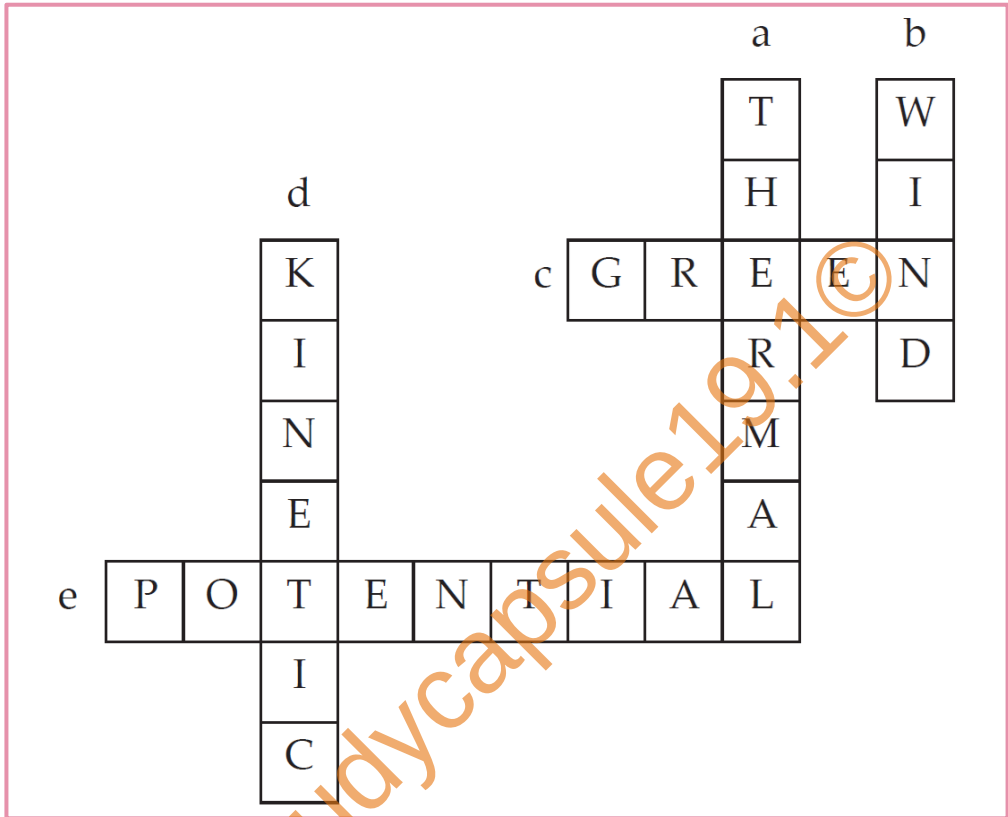


- iv. Stages in electricity generation using wind energy:



5. Solve the following crossword puzzle.

- a. Maximum energy generation in India is done using _____ energy.
- b. _____ energy is a renewable source of energy
- c. Solar energy can be called _____ energy.
- d. _____ energy of wind is used in wind mills.
- e. _____ energy of water in dams is used for generation of electricity.



6. Explain the difference.

- a. Conventional and Non-conventional Sources of energy.

Conventional energy sources	Non-conventional energy sources
Conventional energy sources are exhaustible.	Non-conventional sources of energy are inexhaustible
Conventional sources of energy cause pollution	Non - conventional sources of energy are pollution free.

Conventional sources are used extensively by us.	Non-conventional sources of energy are not used as extensively as conventional sources.
These sources require thousands of years to form.	They are abundantly available in nature.
<u>Examples</u> : Coal, Crude oil, Natural gas, etc.	<u>Examples</u> : Solar energy, wind energy, bio-gas etc.

b. Thermal electricity generation and solar thermal electricity generation.

Thermal electricity generation	Solar thermal electricity generation
In thermal power plant coal, natural gas, etc. are used as sources (fuels).	In solar power plant, solar radiations are used as the source
Thermal power plant causes air pollution.	Solar thermal power plant does not cause air pollution.
The rate of electric power generation is very high.	The rate of electric power generation is very low.
-	-friendly.

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7. What is meant by green energy? Which energy sources can be called as green energy sources and why? Give example.

- i. Green energy means eco-friendly form of energy which does not cause environmental problems and are non-exhaustible, perpetual and sustainable.
- ii. Solar energy, wind energy, hydro electric energy do not harm the environment as there is no combustion involved and there is no air pollution caused.
- iii. Thus, energy obtained from these sources are called as green energy.

8. Explain the following sentences.

a. Energy obtained from fossil fuels is not green energy.

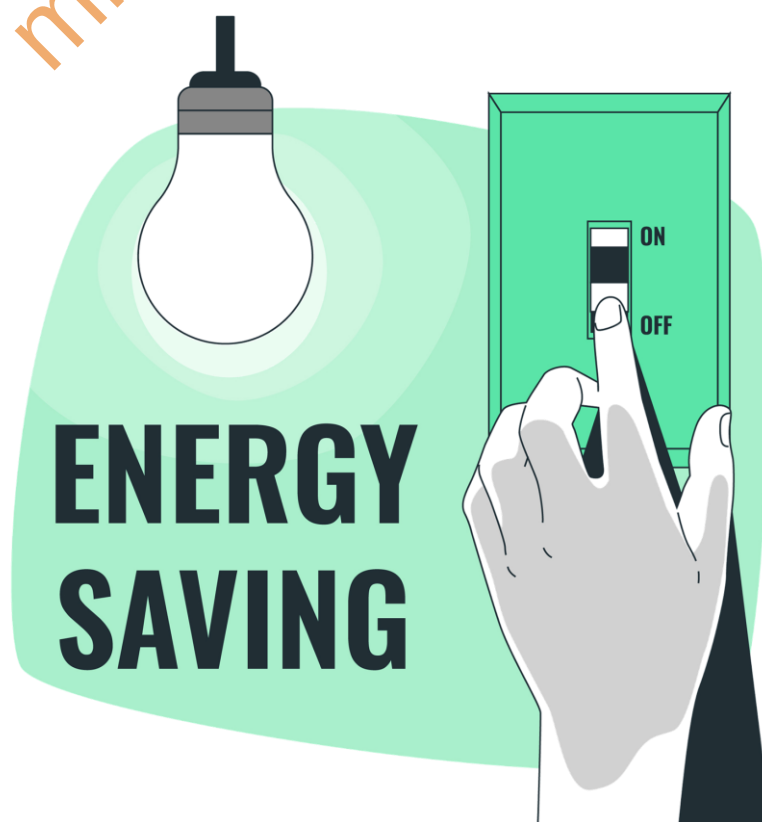
- i. Green energy means eco-friendly form of energy which does not cause environmental problems and are non-exhaustible, perpetual and sustainable.
- ii. Fossil fuels cause pollution during combustion. Burning of fossil fuels results in emission of gases like carbon dioxide, sulphur oxide and nitrogen oxide which are harmful to the health.
- iii. Along with the emission of gases due to burning of fossil fuels, soot particles are also released into the environment. This may cause serious health problems related to the respiratory system.
- iv. The reserves of fossil fuel are limited. Therefore, in future, there will be limitations on their availability.

- v. Thus, due to all such reasons energy obtained from fossil fuels is not green energy.

b. Saving energy is the need of the hour.

Saving energy is the need of the hour because:

- i. In modern civilization, continuous energy supply is needed for the technology and development.
- ii. The energy has become a basic need for man. Energy generation, needs various fuels. The coal and fossil fuels are limited.
- iii. Due to over-exploitation, these reserves are getting fast depleted.
- iv. Use of fossil fuels also resulting in pollution and climate change.
- v. Nuclear energy can be very hazardous.
- vi. A lot of research is being done in the field of green energy, but the tremendous human population always needs more energy.
- vii. Therefore, every person should save the energy, as saving energy is the need of the hour.



9. Answer the following questions.

a. How can we get the required amount of energy by connecting solar panels?

- i. A silicon solar cell of dimension 1 cm^2 generates a current of about 30 mA and potential difference of about 0.5 V. Thus, a silicon solar cell of dimension 100 cm^2 will generate about 3 A ($30 \text{ mA} / \text{cm}^2 \times 100 \text{ cm}^2 = 3000 \text{ mA}$) current and 0.5 V.
- ii. If solar cells are connected in series the potential difference gets added but current remains the same. If solar cells are connected in parallel the potential difference remains the same but the current generated is the summation of current from individual cells.
- iii. By using this principle, we can obtain power of desired quantity.
- iv. We can decide the number of solar panels to be used and what type of combination we can have to get required amount of electric power.
- v. Hence, we can get required amount of energy by connecting solar panels.



b. What are the advantages and limitations of solar energy?

Advantages of solar energy:

- i. Solar energy is green energy i.e., it is renewable and does not cause pollution.
- ii. Solar panels for the generation of electricity requires less maintenance.
- iii. With the help of solar energy, electricity is possible to be generated even in the most remote, inaccessible locations where electric power lines cannot be laid.

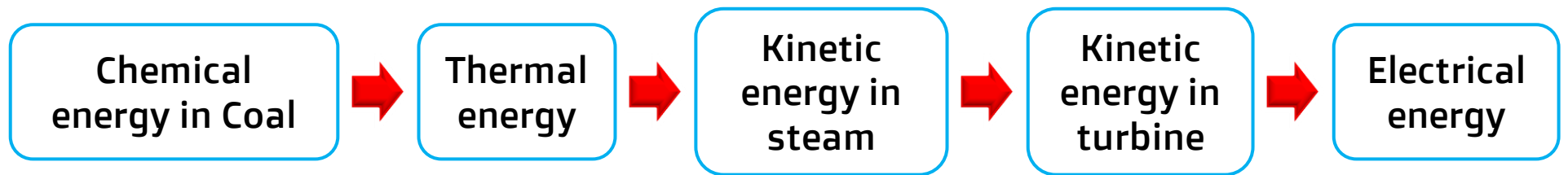
Limitations of solar energy:

- i. Electricity generation based on solar energy is weather dependent and hence it is less reliable. In winters and in cloudy days, the production becomes less.
- ii. The whole set-up of the panels requires lot of space to generate considerable amount of electricity.
- iii. The initial cost of a solar panel is sufficiently high.



10. Explain with diagram step-by-step energy conversion in

a. Thermal power plant



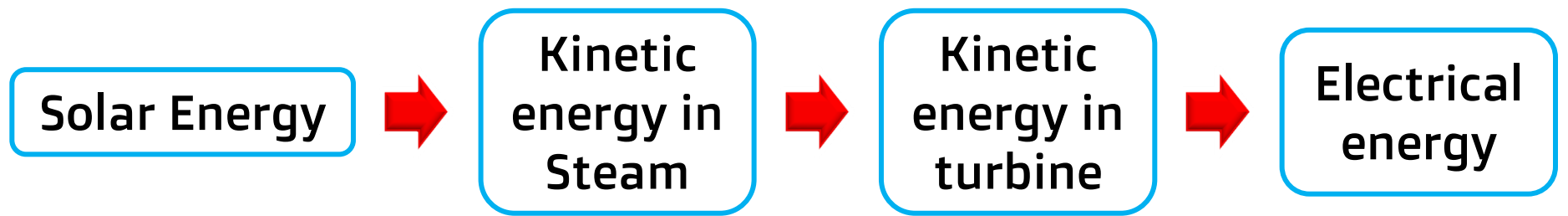
- i. Coal has stored chemical energy, and it is used as a fuel.
- ii. Coal is burnt to obtain heat energy. This is used to boil water to obtain steam.
- iii. Steam of very high temperature and pressure is generated.
- iv. The kinetic energy of the steam drives the turbines thus the generator connected to the turbines rotates and electricity is generated.

b. Nuclear power plant



- i. When fuel U-235 is bombarded with a neutron it becomes unstable and splits into Barium & Krypton and 3 neutrons with a lot of energy (200 MeV).
- ii. By using this energy, water is boiled to form steam.
- iii. By using kinetic energy of steam the turbine is rotated.
- iv. Then the kinetic energy of the turbine is used to rotate the generator and electricity is produced.

c. Solar thermal power plant



- i. Solar energy is initially converted into heat by concentrating it.
- ii. The absorbed sunlight is utilized to boil water and steam is obtained.
- iii. Kinetic energy of steam is used to drive turbines.
- iv. The kinetic energy of turbine is used to drive generators which produce electricity.

d. Hydro - electric power plant



- i. Water stored in a dam's reservoir has potential energy.
- ii. When this water flows, the potential energy gets converted into Kinetic energy.
- iii. With the help of this Kinetic energy, the turbines are rotated.
- iv. The turbine in turn drives the generator to generate electricity.

11. Give scientific reasons

a. The construction of turbine is different for different types of power plants.

- i. Every power plant uses a different kind of source to rotate the turbine.
- ii. In thermal and nuclear power plants, steam produced rotates the turbine.
- iii. In hydroelectric power plant, flowing water is used to rotate turbines and wind is the medium in windmills.
- iv. Also, the power output of every power plant is different.
- v. Hence, there should be turbines of different designs for different power plants.

b. It is absolutely necessary to control the fission reaction in nuclear power plants.

- i. Nuclear power plants work on the principle of controlled chain reaction.
- ii. When slow moving neutrons strike the uranium nucleus, it splits into Barium and Krypton with release of three neutrons and tremendous amount of energy (200 MeV).
- iii. The three neutrons generated in this process cause fission of three other Uranium - 235 atoms releasing more energy.
- iv. If not controlled then this exponential increase in energy can be very dangerous, this concept is harnessed in production of nuclear

bombs.

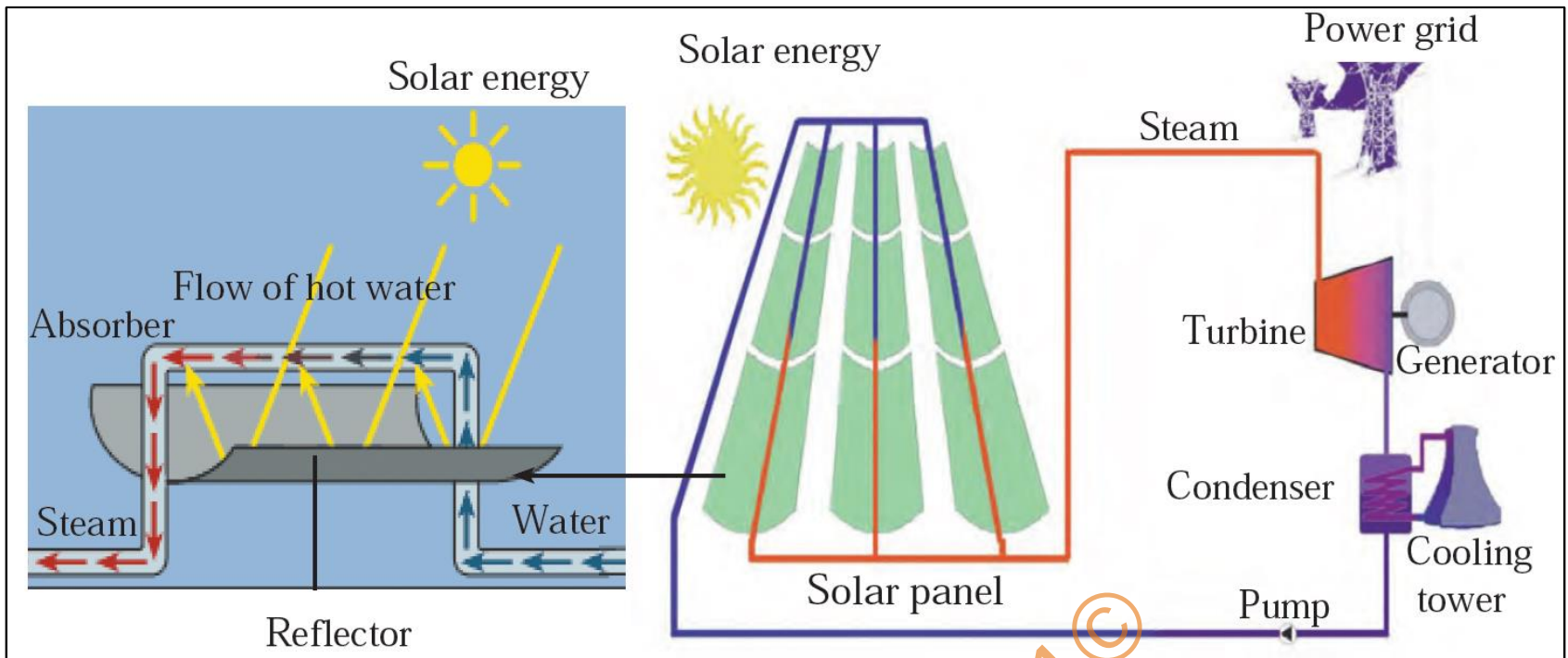
- v. Reaction is controlled by using neutron absorbent material like boron steel and cadmium rods which absorb excess neutrons and heat generated can be used for welfare of mankind.
- vi. Hence, it is necessary to control the fission reaction in nuclear power plants.

c. Hydroelectric energy, solar energy and wind energy are called renewable energies.

- i. Hydroelectric energy, solar energy and wind energy are all obtained without burning of fossil fuel.
- ii. They do not cause air pollution and are thus, eco-friendly.
- iii. Sunrise, rainfall, blowing of wind are natural phenomena.
- iv. These phenomena are constantly and continuously happening on earth.
- v. By setting up hydroelectric power plant, solar panels and wind mills we can obtain these energies continuously, uninterrupted and at low cost.
- vi. Hence, these sources are called renewable energy sources.

d. It is possible to produce energy from mW to MW using solar photovoltaic cells. – Same answer as Q-9-a

12. Draw a schematic diagram of solar thermal electric energy generation.



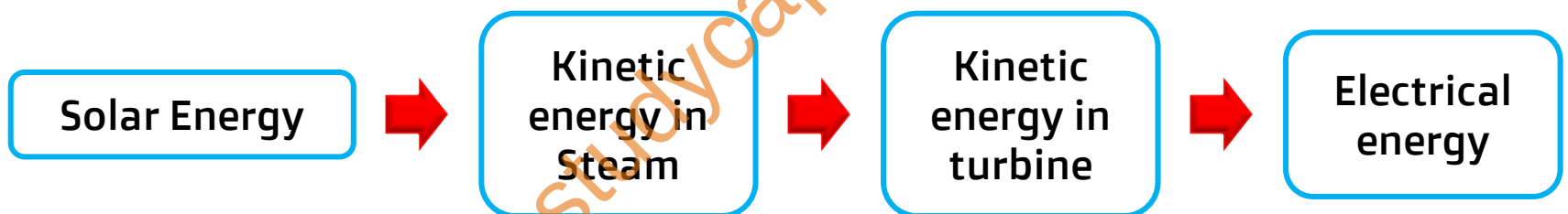
13. Give your opinion about whether hydroelectric plants are environment friendly or not?

- i. Hydroelectric plants have advantages in some respects, but they also create problems in other aspects.
- ii. Since no fuel is burnt in hydroelectric power generation, there is no pollution resulting from combustion of fuels.
- iii. If there is sufficient water storage in the dam, it is possible to generate electricity as and when necessary.
- iv. Although water reservoir is used for power generation, it can be replenished during rainy season leading to uninterrupted power generation.
- v. All these facts might give the impression that hydroelectric power generation is eco-friendly, but it isn't entirely so.

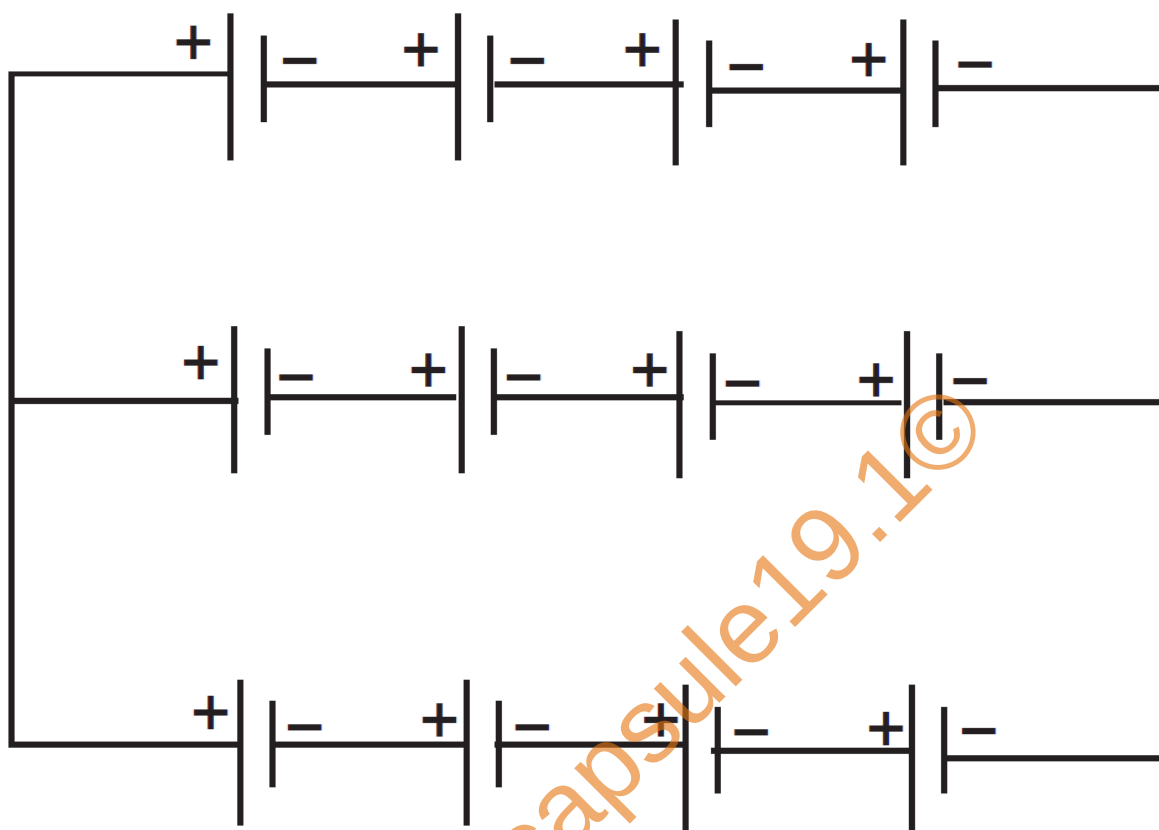
- vi. When a dam and reservoir are constructed, many villages and settlements get submerged.
- vii. Even though displaced people receive resettlement, it often causes a lot of emotional trauma.
- viii. Biodiversity is also affected as forest lands are submerged.
- ix. Dams obstruct the flow of rivers, impacting the aquatic organisms living in the water.
- x. Some people worry that the excessive pressure of water on the land makes the region more prone to earthquakes.

14. Draw neat and labelled diagrams.

a. Energy transformation in solar thermal electric energy generation.



- b. One solar panel produces a potential difference of 18 V and current of 3A. Describe how you can obtain a potential difference of 72 Volts and current of 9 A with a solar array using solar panels. You can use sign of a battery for a solar panel.



15. Write short note on

Electrical energy generation and environment.

- i. The environment impact of electricity generation is significant because modern society needs large amount of electrical power.
- ii. The power is normally generated at power plants that converts some other kind of energy into electrical power.
- iii. Each system has some advantages and disadvantages, but many of them pose environmental concerns.
- iv. Thermal power plants give out CO_2 , NO_2 , SO_2 which causes air pollution.

- v. Nuclear power plant pose the threat of disposal of nuclear waste and radiation leakage.
- vi. Windmills and solar power plants are eco-friendly, but installation cost is high hence is used less frequently.
- vii. Hydroelectric power plants are Eco-friendly and generate sufficient electricity, but rehabilitation of people is necessary, whose land is used for construction of dam.



INTEXT QUESTIONS

1. What is Energy?

Energy is the capacity or stored ability of a body to do work.

2. What are different types of Energy?

There are only two basic types of energy: potential energy and kinetic energy.

3. What are different forms of Energy?

Following are the forms of energy :

(i) Mechanical energy (ii) Heat energy (iii) Light energy (iv) Sound energy
(v) Electrical energy (vi) Magnetic energy (vii) Chemical energy (viii)
Nuclear energy

4. Make a table based on forms of energy and corresponding devices.

Forms of Energy	Devices
a. Electric	Electric Fan, Electric Iron
b. Mechanical	Sewing machine, Bicycle
c. Thermal	Steam Engine, Steam turbine
d. Solar	Solar cooker, Solar heater

5. Where do we use electrical energy in our day-to-day life?

We use electric energy in various appliances such as bulbs and tube lights, fans, iron, refrigerator, water pumps, cellphone chargers, television sets, tape recorders and computers.

6. How Electric energy is produced ?

- i. Most of the electric power plants are based on the principle of electro - magnetic induction invented by Michael Faraday.
- ii. According to this, whenever magnetic field around a conductor changes a potential difference is generated across the conductor and electrical energy is produced.

7. Why the energy in the coal is called as chemical energy?

- i. Coal contains carbon and hydrocarbons which produce heat on combustion.
- ii. Hence energy in coal is called chemical energy.

8. Why is steam used to rotate the turbine?

- i. The steam is used to rotate the turbine because the steam is the vapourised water that can develop very high kinetic energy required to rotate the turbine.
- ii. Steam can be converted back to water and re-circulated for usage.
- iii. Also, water is an easily available fluid and never harms the environment thus steam is preferred to rotate the turbine.

9. How does nuclear fission take place?

- i. When a slow-moving neutron is bombarded on Uranium - 235 nucleus, it absorbs the neutron and gets converted into its isotope Uranium - 236.
- ii. Uranium - 236 being extremely unstable, converts into atoms of Barium and Krypton through a process of fission releasing three neutrons and 200 MeV energy.

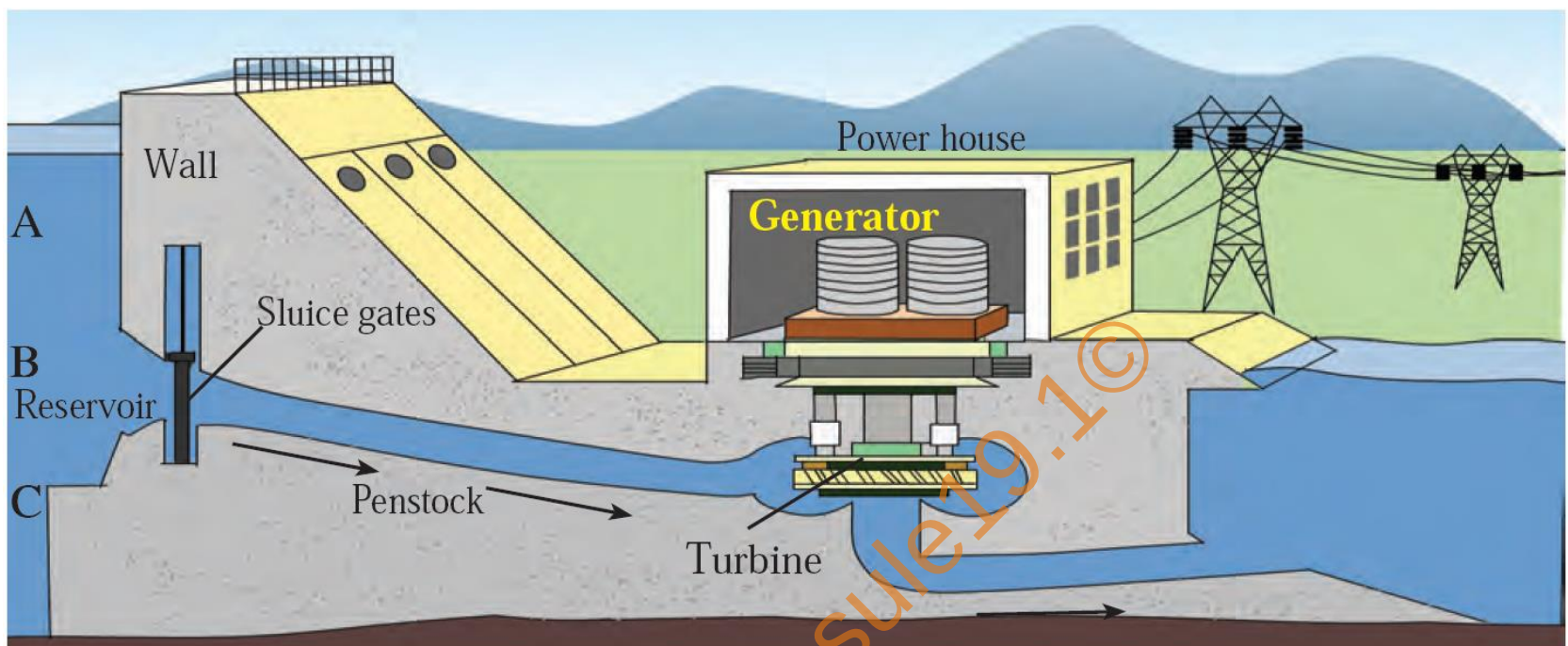
Tamil Nadu	1863
Maharashtra	1280
Rajasthan	995
Karnataka	808

10. Which electricity generation process is eco-friendly and which not?

- i. Electricity generated through solar energy and wind energy is truly eco-friendly.
- ii. Although it is said that hydroelectricity is non-polluting and eco-friendly, this is not true.

- iii. Hydroelectric projects cause the destruction of biodiversity and displacement of local people.
- iv. Thermal energy, nuclear energy, and energy obtained through natural gas are not eco-friendly at all.

11. Use your brain power



- I. **With reference to point B, potential energy of how much water reservoir in the dam will be converted into kinetic energy?**

The potential energy of water stored in the region A to B, will be converted into kinetic energy.

- II. **What will be the effect on electricity generation, if the channel taking water to turbine starts at point A?**

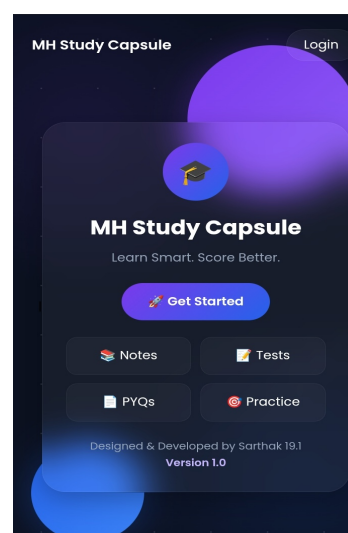
The energy generation will increase as the water at A It will have higher potential energy that can generate higher kinetic energy as it flows.

- III. **What will be the effect on electricity generation, if the channel taking water to turbine starts at point C?**

The energy generation will decrease as the water at C will have lower potential energy that will generate lower kinetic energy as it flows.

12. What is lake tapping? Why it takes place?

- i. A lake tap involves excavating a tunnel almost to the water or rock contact and then blasting out the final protective rock to allow water to suddenly inflow into the tunnel from the lake.
- ii. The first lake tapping was carried out at Koyana dam 13 years ago. Apr 26, 2012. This process allows the lake water to flow down from that tunnel. This type of lake tapping was first established in the Asian continent.
- iii. Lake tapping is done for various reasons :
 - a. Limit the lake water level
 - b. Depollute the lake water
 - c. Drain out the lake water effectively
 - d. Provide waterways to establish hydropower
 - e. Make water available for irrigation



13. Observe the schematic of thermal power plant and the nuclear power plant. Discuss what are the similarities and differences between the two?

Similarity: In both the power plants water is boiled to form steam and kinetic energy of the steam rotates the turbine.

Differences: In the thermal power plant coal is a main fuel that is burnt. In the nuclear power plant Uranium - 235 is the fuel which undergoes nuclear fission.

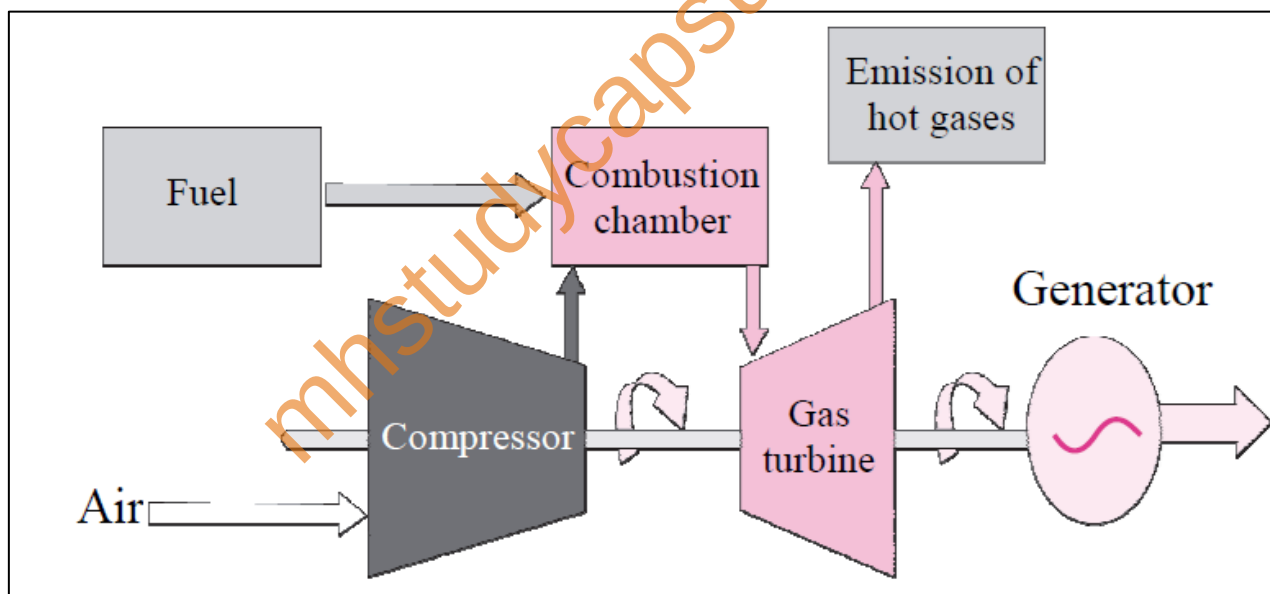
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Previous Year Questions

1. What are the advantages of hydroelectric power generation?

- Since no fuel is burnt in hydroelectric power generation, there is no pollution resulting from combustion of fuels.
- If there is sufficient water storage in the dam, it is possible to generate electricity as and when necessary.
- Although water reservoir is used for power generation, it can be replenished during rainy season leading to uninterrupted power generation.

2. Observe the diagram and answer the questions:



a. Which energy is generated from the power plant?

Electrical energy

b. State its source.

Natural gas

c. Which is more ecofriendly? Power generation from coal or Power generation from natural gas? Why?

Power generation from natural gas is more eco-friendly.

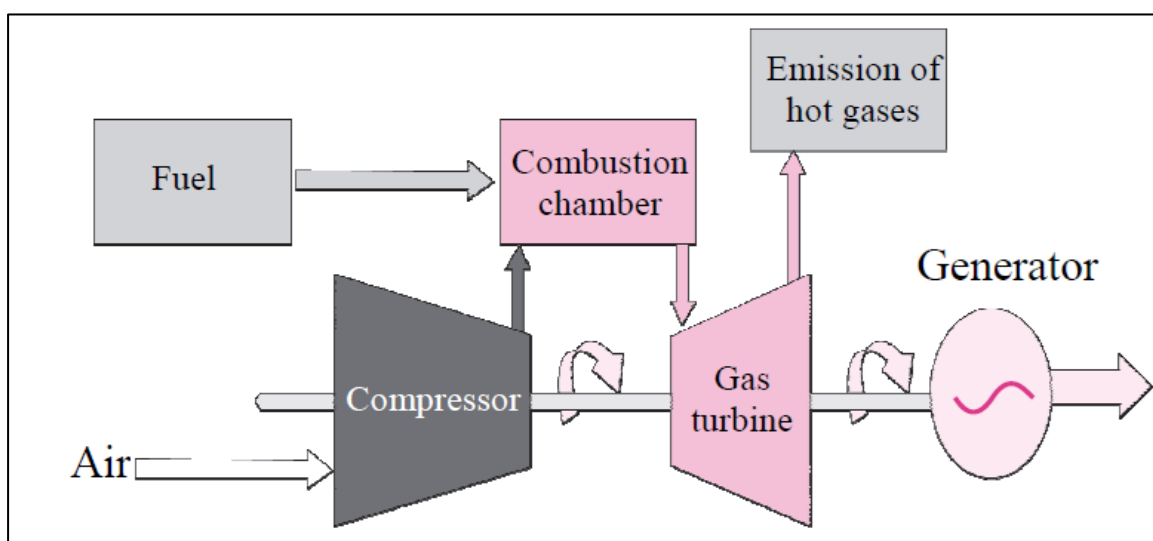
The efficiency of this type of power generation plant is higher than that of power generation plant based on coal. Moreover, since the natural gas does not contain sulphur, burning of natural gas results in less pollution.

3. Hydroelectric energy, solar energy and wind energy are called renewable energies. Why? – Exercise Question 11-c

4. Draw schematic of power plant based on natural gas and answer the following questions:

- At which place natural gas power plant is situated in Maharashtra?
- How is pollution reduced in natural gas-based power plant?
- Give two examples of eco-friendly electricity process.

Answer:



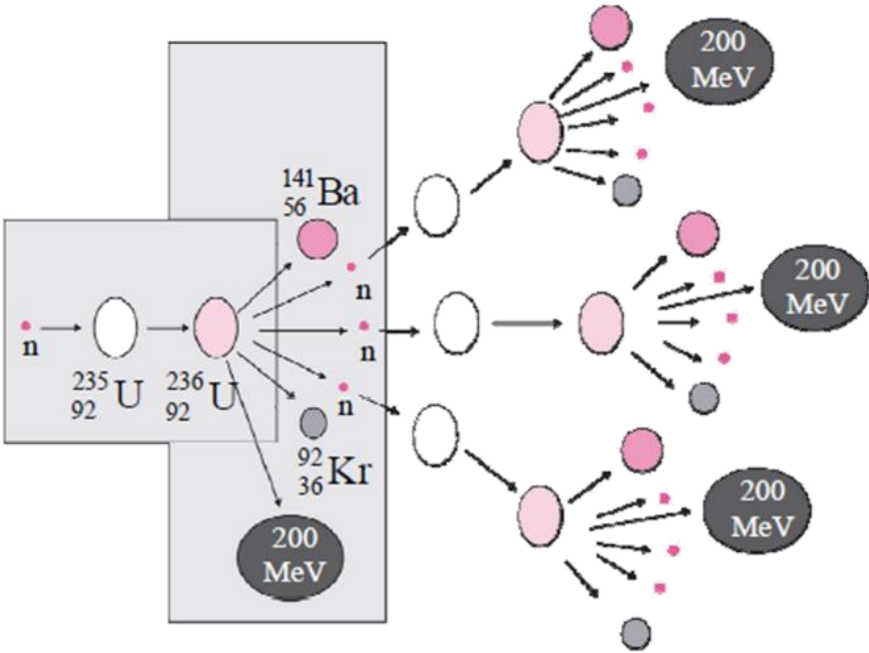
- i. Anjanvel
- ii. Since the natural gas does not contain sulphur, burning of natural gas results in less pollution.
- iii. Solar energy, wind energy

5. Make a table based on forms of energy and corresponding devices:

	Forms of Energy	Devices
a.	Electric	_____
b.	Mechanical	Sewing machine, Bicycle
c.	Thermal	_____
d.	_____	Solar cooker, Solar heater

Forms of Energy	Devices
a. Electric	Electric Fan, Electric Iron
b. Mechanical	Sewing machine, Bicycle
c. Thermal	Steam Engine, Steam turbine
d. Solar	Solar cooker, Solar heater

6. Answer the following questions with respect to the below figure.

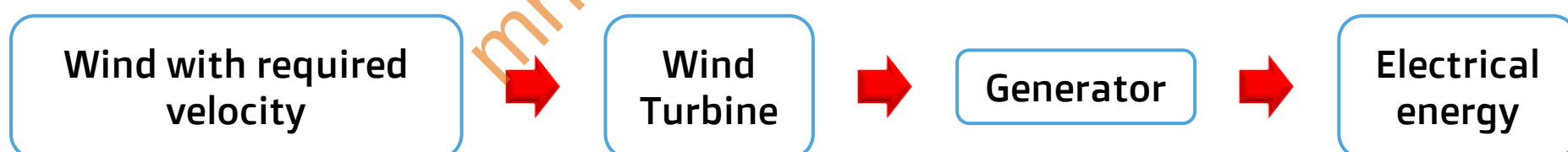


- a. **Identify the reaction shown in the figure. – Nuclear fission**
- b. **Where is this reaction used? – This reaction is used in nuclear power plant.**
- c. **Which atom is used in this reaction? – Uranium - 235**

7. Power generation plant based on natural gas is eco-friendly. Give reason.

- i. Power generation plant based on natural gas is eco-friendly because natural gas contains no sulphur. As a result, burning produces less pollution.
- ii. Natural gas emits 50 to 60 percent less carbon dioxide (CO_2) in power plants than regular oil or coal-fired power plants.
- iii. It also emits greenhouse gases into the atmosphere with a shorter life cycle. As a result, it is environmentally friendly.

8. Complete the stages in electric generator using wind energy.



Write limitations in an electric generator using wind energy.

- i. The wind's speed is not constant, and the required wind velocity is not available everywhere.
- ii. Wind power has the limitation of producing no electricity when the wind is not blowing.
- iii. Installation of wind energy farm is expensive.

9. It is absolutely necessary to control the fission reaction in nuclear power plant. – **Exercise Question 11 - b**

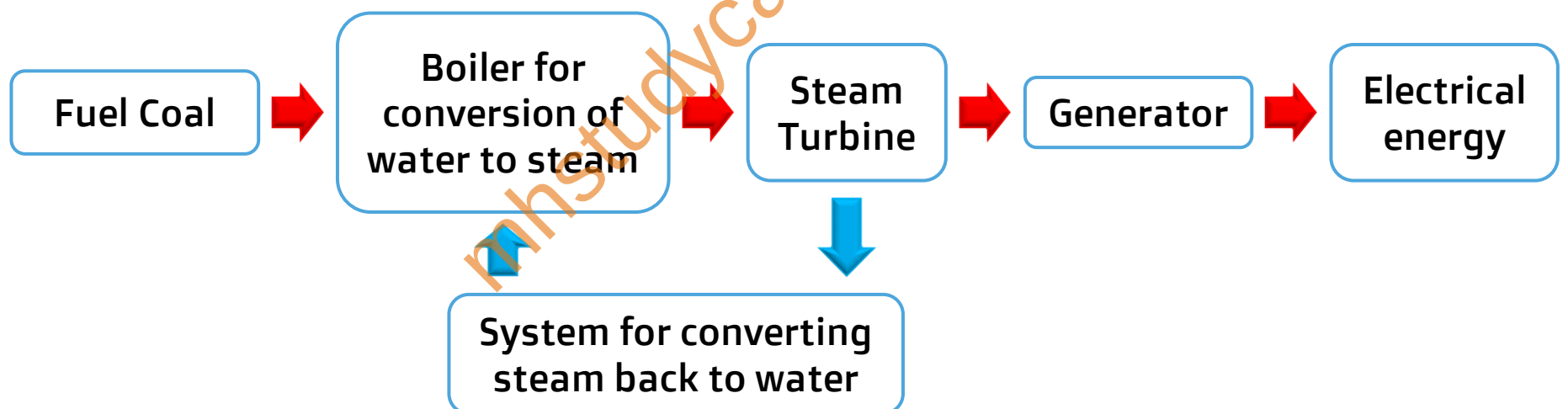
10. Answer the following questions from figure: - **Intext Question**

(a) With reference to point B, potential energy of how much water reservoir in the dam will be converted into kinetic energy?

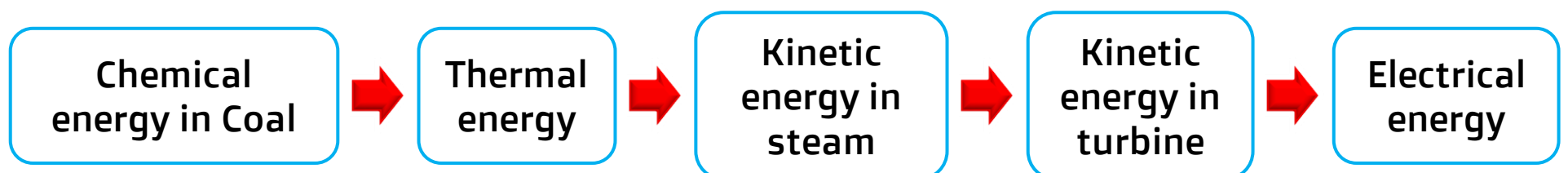
(b) What will be the effect on electricity generation, if the channel taking water to turbine starts at point A?

(c) What will be the effect on electricity generation, if the channel taking water to turbine starts at point C?

11. Complete the given flow chart for electrical energy production based on thermal power plant.



Complete the flow chart for energy transformation in thermal power plant.



12. Write any two advantages and two problems associated with hydroelectric power plant.

Advantages of hydroelectric power generation

1. Since no fuel is burnt in hydroelectric power generation, there is no pollution resulting from combustion of fuels.
2. If there is sufficient water storage in the dam, it is possible to generate electricity as and when necessary.
3. Although water reservoir is used for power generation, it can be replenished during rainy season leading to uninterrupted power generation.

Problems associated with hydroelectric power plant

1. The back-water due to storage of water in dam may submerge villages or towns in that area. This leads to the problems of rehabilitation of the displaced population. Moreover, this can also submerge forests as well as fertile land.
2. The obstruction of the flow of river water may have adverse effect on living world in the river.

Is hydroelectric power generation environment friendly? What is your opinion? – Exercise Question 13

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